

BeautifulSoup Documentation Analytics System

Final Analytical Report — Collect · Parse · Extract · Analyze · Visualize

Sections extracted: 113 | Links extracted: 504 | Code examples extracted: 220

1. Dataset Overview

The dataset was built from a single technical documentation page: the official BeautifulSoup 4 documentation (<https://www.crummy.com/software/BeautifulSoup/bs4/doc/>). The page was downloaded once, parsed, and broken down into three structured tables covering sections, hyperlinks, and Python code examples.

Dataset	Rows	Description
sections.csv	113	Headings (h1/h2/h3) with text, word count, code block count, link count
links.csv	504	All hyperlinks with text, href, type, and originating section
code_examples.csv	220	Python code blocks with line count and BS4-method usage flags

2. Scraping Method

- requests was used to send an HTTP GET request to the target URL and verify the HTTP status code before saving.
- The raw HTML response was saved to data/raw/beautifulsoup_doc.html for reproducibility.
- BeautifulSoup parsed the saved HTML using Python's built-in html.parser. The official documentation also describes lxml as a faster alternative and html5lib as a more lenient but slower alternative.
- Sections were identified via h1/h2/h3 headings; text between one heading and the next was assigned to that section.
- Links were extracted by iterating over soup.find_all('a') and reading each href attribute, then classified into 5 categories.
- Code examples were extracted from tags and analyzed for the presence of key BeautifulSoup methods (find_all, find, select, get_text) and the requests library.

3. Extracted Data Summary

A total of 113 documentation sections were identified. The longest section by word count is "Multi-valued attributes" with 852 words. Link classification found 470 internal anchors and 19 external links, among other link types recorded in links.csv.

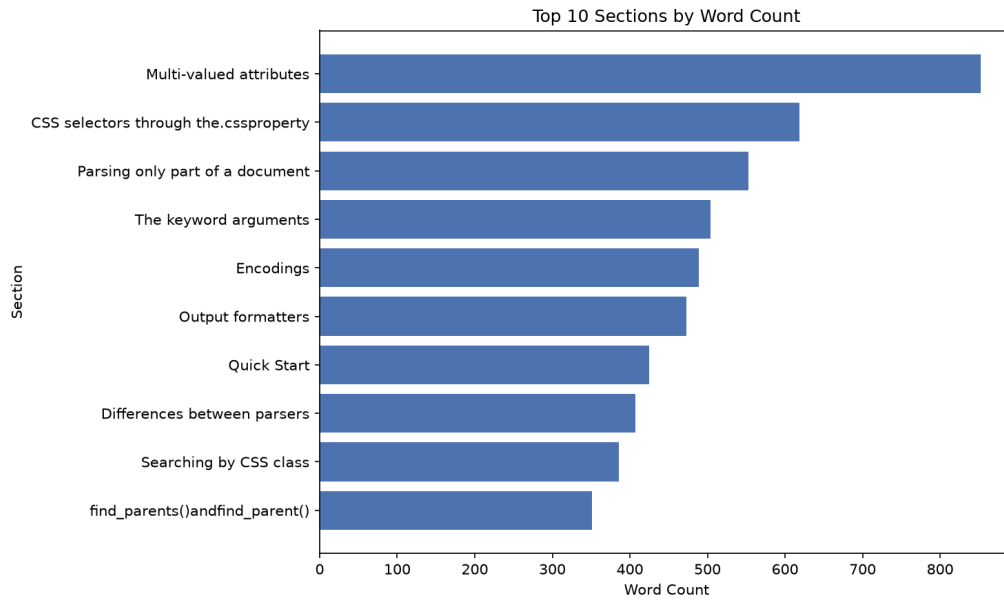
4. Analysis Results

(* = additional question proposed by the team)

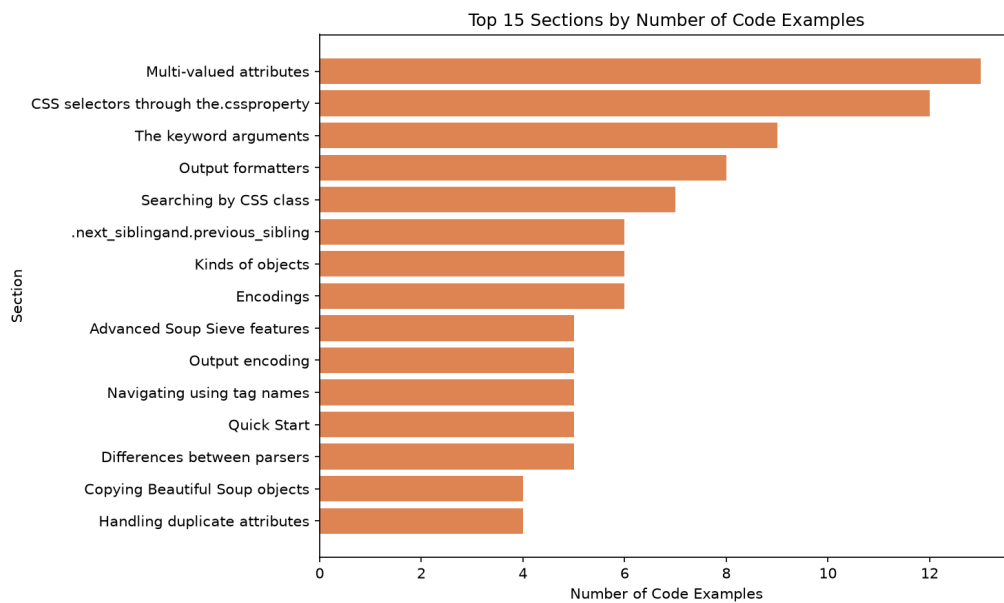
#	Question	Answer
Q1	How many sections are in the documentation?	113
Q2	Which section has the highest word count?	Multi-valued attributes (852 words)
Q3	Which section contains the most code examples?	Multi-valued attributes (13)
Q4	Which section contains the most links?	Table of Contents (128)
Q5	Top 10 most frequent technical keywords	soup, tag, html, class, find, string, link, href, parser, beautifulsoup
Q6	Internal vs. external links	internal: 470, external: 19
Q7	Code examples using find_all()	39
Q8	Code examples using get_text()	4
Q9*	Word count statistics (mean / median / std)	173.4 / 139.0 / 134.8
Q10*	% of code examples using a BS4 search method	30.0%

5. Charts

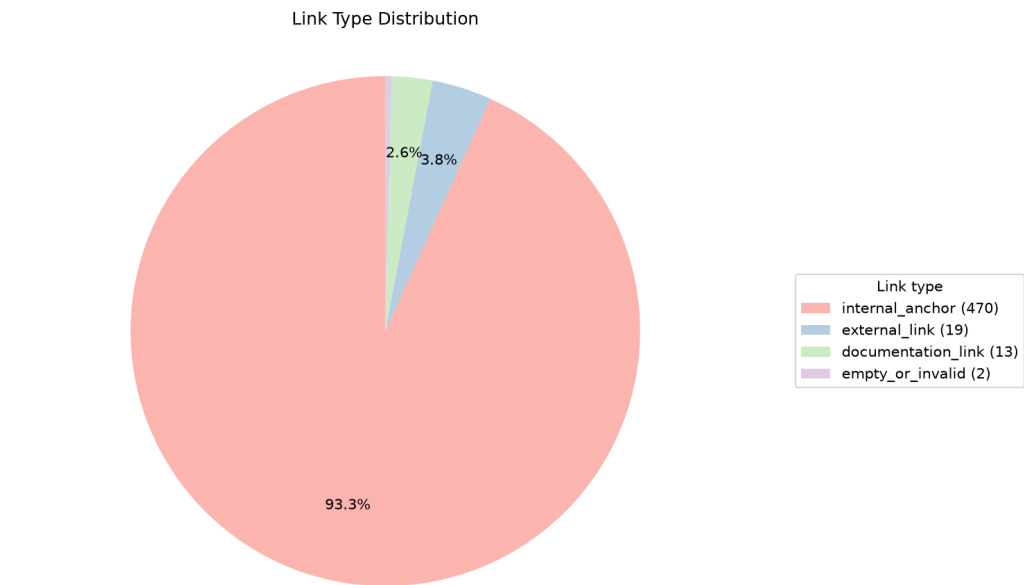
Top 10 Sections by Word Count



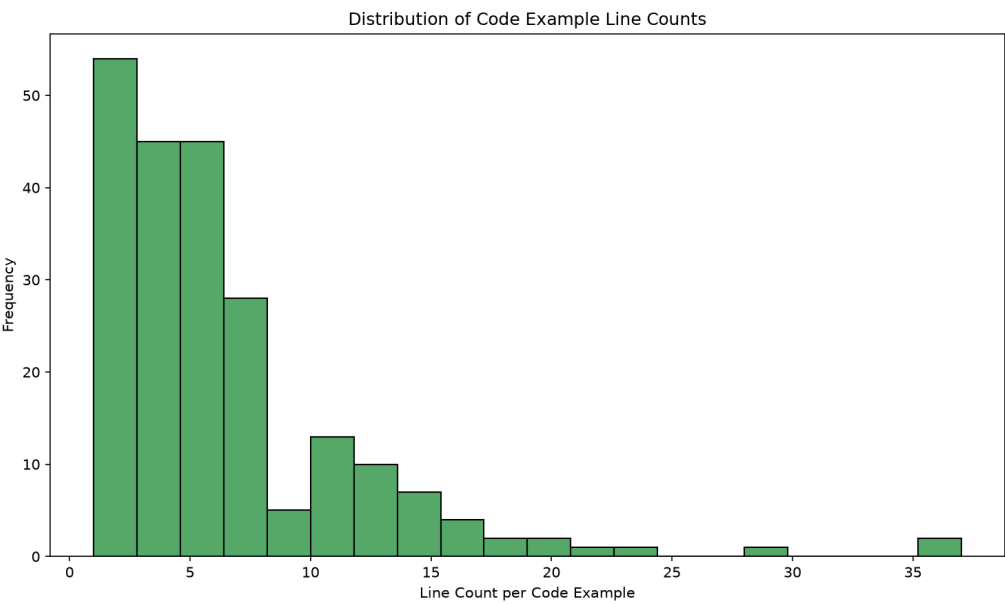
Number of Code Examples by Section



Link Type Distribution



Code Example Line Count Distribution



6. Key Findings

- The section “Multi-valued attributes” is the most content-heavy part of the documentation, suggesting it is a core reference area for users.
- The majority of links (470 out of 504) are internal anchors, meaning the documentation is largely self-contained and relies on in-page navigation rather than external references.
- Only 39 code examples explicitly use `find_all()`, while 30.0% of all code examples use at least one core search method (`find_all/find/select`), showing these methods are central but not universal across all examples.
- `get_text()` appears in only 4 code examples, indicating it is a less emphasized method compared to element search methods.
- Section word counts vary widely ($\text{std} \approx 135$ words), reflecting a mix of short reference sections and long explanatory ones.

7. Limitations

- The analysis is based on a single snapshot of the documentation page; results will change if the page is updated.
- Keyword extraction uses a simple tokenizer and a minimal stopword list, so some non-technical words may still appear in the top-10 list.
- Sections are delimited purely by heading tags (`h1/h2/h3`); pages with inconsistent heading structure could lead to mis-assigned content.
- Link classification relies on hostname matching for `'documentation_link'` vs `'external_link'`, which may misclassify uncommon URL formats.
- The system currently analyzes only one documentation page; it has not been tested for scale across multiple pages or sites.

8. Conclusion

This project successfully built an end-to-end pipeline that collects, parses, extracts, analyzes, and visualizes the Beautiful Soup documentation. Across 113 sections, 220 code examples, and 504 classified links, the analysis shows a documentation page that is text-heavy, strongly self-referential through internal anchors, and centered on a small set of core BeautifulSoup search methods. The pipeline is modular (collector, parser, extractor, analyzer, visualizer, report generator) and can be re-run end-to-end whenever the source documentation changes.